

Leo J. Paggen

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Education

M.S. Data Science

Maastricht University, Netherlands

Sep 2024 – Sep 2026

- **Relevant Coursework:** Machine Learning, Statistical Modeling, Optimization, Natural Language Processing, Algorithms for Big Data, Time Series Forecasting

Experience

Red2Green - Research Intern

Rotterdam, NL

April 2026 – Sep 2026

- Master's thesis internship

Maastricht University - Teaching Assistant

Maastricht, NL

Jan 2025 – Jul 2026

- Supported courses in Natural Language Processing, Principles of Programming Languages, and Linear Programming, mentoring undergraduate students in ML, optimization, debugging, and data science projects.

PwC - Data Consultant Intern

Brussels, Belgium

Nov 2025 – Jan 2026

- Developed an AI-assisted classification pipeline mapping product descriptions to 4-digit Combined Nomenclature codes for EU Deforestation Regulation compliance.
- Evaluated embedding retrieval and frontier LLM approaches, selecting the strongest architecture through comparative experiments.
- Designed a two-stage LLM validation pipeline that improved recall from 80% to 88% on internal evaluation data.

Amazon - Data Scientist Intern

Seattle, WA

Jul 2025 – Oct 2025

- Extended a production demand forecasting system supporting every North American fulfillment operations.
- Owned three research initiatives in uncertainty estimation, recurring event modeling, and operational disruption modeling, evaluating improvements using Weighted Absolute Percentage Error (WAPE).
- Built a bootstrap uncertainty framework with multiprocessing, achieving up to 4.7× speedup, and developed holiday and yard-closure forecasting overlays reducing WAPE by up to 2% in testing conditions.
- Extended Redshift/SQL data pipelines, authored a 10-page internal research paper, and presented findings to scientists, engineers, and non-technical stakeholders.

Personal Projects

Static Verification Framework for PyTorch Tensor Programs

Python, Rust, Z3, Protobuf

- Static type checker for Python written in Rust using a custom translation library. The tool uses Z3 SMT Solver to propagate constraints as it walks the program and informs the user of any errors before runtime.
- Branch-conditioned tensor types are retained across control flow graph joins and subsequently analyzed to discover infeasible paths.

LLM-Based Maritime Autonomy Platform

Python, vLLM, Qwen

- Designing an multi-LLM-assisted autonomy layer for MOOS-IvP that combines natural-language reasoning with deterministic vessel control.
- Each LLM owns one responsibility; one is using RAG to retrieve regulations, one uses RAG to retrieve similar historic incidents, and the other models infer based on deterministic information to produce natural language.
- The end result is a multi-agent model that can read its environment and pilot a boat from point A to B safely in the simulation.

Cisco ACIBILIZER

Python, Ansible

- Developed an open-source tool that converts Cisco ACI configurations into reproducible Ansible automation.

Technical Skills

Languages: Python, Rust, Java, SQL

Machine Learning: Time Series Forecasting, Statistical Modeling, LLMs, NLP, Supervised Fine-Tuning, Uncertainty Quantification

Systems & Tools: PyTorch, vLLM, Z3 SMT Solver, Static Program Analysis, Protobuf, Redshift, Git